

Problem Pack

Five Suggested Directions for the ODE IVP Project

Each direction below gives the phenomenon, mathematical model, numerical challenges, and specific questions for the report. The deterministic ODE directions use the usual requirements of three numerical methods, stability, adaptivity, and validation. Direction ④ is an SDE extension pathway: replace deterministic stability/adaptivity analysis by exact–EM–Milstein comparison for GBM, strong and weak convergence with coupled paths, Monte Carlo uncertainty, and nested-grid validation for the non-affine model. In every direction, follow its topic-specific tasks where they specialise the generic project brief.

Pick one and defend the choice in one paragraph in your Week-1 milestone entry. (You may also propose your own topic — state it in your Week-1 milestone entry.)

1. ① Celestial mechanics: restricted three-body problem

Phenomenon. A small body moves under the gravity of two much heavier bodies (e.g. spacecraft in the Earth–Moon system). The restricted three-body problem (RTBP) is the standard model used for mission design to Lagrange points. The model and its equations below follow the standard formulation; classical references are Szebehely (1967) *Theory of Orbits: The Restricted Problem of Three Bodies* (Academic Press) and Koon, Lo, Marsden & Ross (2006) *Dynamical Systems, the Three-Body Problem and Space Mission Design* (free at <http://www.cds.caltech.edu/~marsden/books/>) — the historical setup goes back to Euler and Lagrange (1772).

Model. In the uniformly rotating barycentric frame, the planar circular RTBP in dimensionless units is

$$x'' - 2y' = \partial\Omega/\partial x, \quad y'' + 2x' = \partial\Omega/\partial y,$$

where

$$\Omega = (1 - \mu)/r_1 + \mu/r_2 + \frac{1}{2}(x^2 + y^2), \quad \mu = m_2/(m_1 + m_2),$$
$$r_1^2 = (x + \mu)^2 + y^2 \quad \text{and} \quad r_2^2 = (x - 1 + \mu)^2 + y^2.$$

Or a plain Newtonian N -body system (here $N = 3$: the test particle plus the two primaries moving on their circular two-body orbit)

$$r'' = -G m_1 (r - r_1)/|r - r_1|^3 - G m_2 (r - r_2)/|r - r_2|^3,$$

with $r_1(t), r_2(t)$ the known circular two-body solution (do *not* hold the primaries fixed — that is Euler’s two-fixed-centres problem, a different system). This is the inertial-frame form; it is equivalent to the rotating-frame RTBP above after a change of coordinates.

Write the first-order state as $z = (x, y, v_x, v_y)^T$, with $v_x = x'$ and $v_y = y'$. The equations are singular at either primary ($r_1 r_2 = 0$), so the IVP is considered only up to the first collision unless a regularisation is introduced.

Invariant. The rotating-frame system conserves the *Jacobi integral*

$$C_J = 2\Omega(x, y) - (v_x^2 + v_y^2).$$

It does *not* conserve ordinary mechanical energy and angular momentum separately. In the inertial description the prescribed moving primaries make the test-particle equations explicitly time-dependent, so the test particle’s energy and angular momentum are not separate invariants there either.

Choosing μ . The mass parameter $\mu = m_2/(m_1 + m_2)$ is the only physical parameter left after normalisation:

- Earth–Moon: $\mu \approx 1.215 \times 10^{-2}$ (the classic mission-design value; L_1 and L_2 near the Moon).
- Sun–Earth: $\mu \approx 3.04 \times 10^{-6}$ — much smaller, so the primaries are extremely unequal; L_1/L_2 sit very close to the smaller body.
- Take $\mu \approx 0.5$ for a visually symmetric, two-equal-masses test (good for debugging; the geometry is cleaner but not astrophysical).
- Any $0 < \mu < 1/2$; the equations stay valid, and you can study how $L_1, \dots, L_5$ move as μ varies.

Reproducible baseline. Use the Earth–Moon value $\mu = 0.01215$ and start near the triangular equilibrium,

$$z(0) = (\tfrac{1}{2} - \mu + 10^{-2}, \tfrac{\sqrt{3}}{2}, 0, 0)^T, \quad 0 \leq t \leq 100.$$

Use this common IVP for the required Euler, RK4, and implicit-method comparison. State the norm used for trajectory error and construct a tight-tolerance reference solution. If you choose a different trajectory, report all four initial values, μ , the time interval, and a stopping rule for close approach (for example, terminate when $\min(r_1, r_2) < r_{\min}$).

Numerical challenges. Smooth trajectories that remain away from the primaries are normally accuracy-limited rather than strongly stiff. A close encounter is different: the inverse-distance singularity creates a rapidly varying force and severe step-size restrictions, even though this should not be described automatically as classical stiffness. Monitor C_J and trajectory error separately: a small invariant defect alone does not prove an accurate orbit.

The RTBP has a Hamiltonian formulation, but its rotating-frame Hamiltonian is not the separable $T(p) + V(q)$ form for which the elementary Verlet formula is derived. A symplectic comparison is therefore an *optional advanced extension*: use a method derived for the canonical RTBP Hamiltonian and state the canonical variables and splitting. Symplectic methods generally conserve a nearby modified Hamiltonian and exhibit bounded, oscillatory long-time invariant error; they do not conserve the original energy to machine precision.

Your report should address:

- Verify the expected convergence order of at least two required methods against the common high-accuracy reference on the baseline IVP.
- Plot the defect $|C_J(t) - C_J(0)|$ and trajectory error. Explain why either diagnostic on its own is insufficient.
- Compare the absolute-stability regions with $h\lambda_j$, where λ_j are eigenvalues of the Jacobian evaluated at a stated equilibrium or along the computed trajectory. Determine from evidence whether accuracy or stability is the binding step-size constraint.
- Demonstrate adaptive step-size control on the baseline. A close encounter is an optional stress test: compare work at matched error, enforce a stated $r_{\min}$ event, and do not claim that adaptivity resolves the collision singularity itself.
- *Optional (☆☆☆)*: compare a correctly derived symplectic method with RK4 over a long interval, using matched force evaluations or wall time; report bounded drift, phase/trajectory error, and Jacobi defect rather than claiming exact energy conservation.

Sanity checks. At either triangular equilibrium $z = (\tfrac{1}{2} - \mu, \pm\sqrt{3}/2, 0, 0)^T$, the right-hand side should vanish to round-off. Along every accurately resolved RTBP trajectory, C_J is constant. As a separate code-validation problem, integrate a circular two-body Kepler orbit for one period: the state error should decrease at the method's expected order. For a symplectic method, expect bounded long-time energy error, not machine-precision conservation of the original energy.

2. ② Hodgkin–Huxley neuron model

Phenomenon. The firing of a nerve cell membrane. The Hodgkin–Huxley (HH) model describes how the membrane potential $V(t)$ is driven by gated ion currents. This is the classic electrophysiology model that motivated implicit and adaptive integration methods. At 6.3 °C a single compartment has fast and slow modes whose separation changes markedly during a spike. Explicit methods are usable at sufficiently small steps, but whether accuracy or absolute stability is the active restriction must be established for the specified trajectory and method rather than assumed in advance. Networks and higher-temperature variants can exhibit a substantially larger separation of time scales.

Model. Four coupled ODEs (V , plus gating variables m, h, n):

$$\begin{aligned} C dV/dt &= I_{\text{ext}} - \bar{g}_{\text{Na}} m^3 h (V - V_{\text{Na}}) - \bar{g}_{\text{K}} n^4 (V - V_{\text{K}}) - \bar{g}_{\text{L}} (V - V_{\text{L}}), \\ dm/dt &= \alpha_m(V)(1 - m) - \beta_m(V) m, \\ dh/dt &= \alpha_h(V)(1 - h) - \beta_h(V) h, \\ dn/dt &= \alpha_n(V)(1 - n) - \beta_n(V) n. \end{aligned}$$

Parameters (standard shifted convention). The constants below are the common modern representation of the Hodgkin–Huxley parameter set, with the voltage origin shifted so that rest is near -65 mV. The original 1952 paper used a different voltage origin. Here V is in mV, time is in ms, and current density is in $\mu\text{A}/\text{cm}^2$:

Parameter	Value	Meaning
C	$1 \mu\text{F}/\text{cm}^2$	membrane capacitance
$\bar{g}_{\text{Na}}$	$120 \text{ mS}/\text{cm}^2$	max Na^+ conductance
$\bar{g}_{\text{K}}$	$36 \text{ mS}/\text{cm}^2$	max K^+ conductance
$\bar{g}_{\text{L}}$	$0.3 \text{ mS}/\text{cm}^2$	leak conductance
V_{Na}	$+50 \text{ mV}$	Na^+ reversal potential
V_{K}	-77 mV	K^+ reversal potential
V_{L}	-54.4 mV	leak reversal potential

Rate functions (standard shifted form, V in mV, resting state near -65 mV):

$$\begin{aligned} \alpha_m(V) &= \frac{0.1 (V + 40)}{1 - e^{-(V+40)/10}}, & \beta_m(V) &= 4 e^{-(V+65)/18}, \\ \alpha_h(V) &= 0.07 e^{-(V+65)/20}, & \beta_h(V) &= \frac{1}{1 + e^{-(V+35)/10}}, \\ \alpha_n(V) &= \frac{0.01 (V + 55)}{1 - e^{-(V+55)/10}}, & \beta_n(V) &= 0.125 e^{-(V+65)/80}. \end{aligned}$$

Other textbooks sometimes shift these potentials by a constant (the offset takes the resting potential to 0 mV); the dynamics are identical — pick one convention and state it.

The displayed formulas for α_m and α_n have removable singularities at $V = -40$ and $V = -55$ mV. Do not evaluate them as literal 0/0. Define

$$\psi(x) = \frac{x}{1 - e^{-x}}, \quad \psi(0) = 1,$$

and compute it stably as $x/(-\expm1(-x))$, using a short Taylor expansion near zero. Then

$$\alpha_m(V) = \psi\left(\frac{V + 40}{10}\right), \quad \alpha_n(V) = 0.1 \psi\left(\frac{V + 55}{10}\right).$$

Reproducible benchmark. Work at 6.3°C with the parameters above and

$$V(0) = -65, \quad m(0) = 0.05293, \quad h(0) = 0.59612, \quad n(0) = 0.31768, \quad 0 \leq t \leq 40 \text{ ms}.$$

Apply the rectangular current pulse

$$I_{\text{ext}}(t) = \begin{cases} 20 \mu\text{A}/\text{cm}^2, & 5 \leq t < 6 \text{ ms}, \\ 0, & \text{otherwise.} \end{cases}$$

Treat $t = 5$ and $t = 6$ as event points: end a step at each discontinuity and restart the integrator so that no method averages the jump differently. This complete IVP is the common comparison problem; any alternative stimulus must state its amplitude, onset, duration, temperature, initial state, and final time.

Source. A. L. Hodgkin and A. F. Huxley, “A quantitative description of membrane current and its application to conduction and excitation in nerve,” *Journal of Physiology* **117** (1952), 500–544. The rate-function forms above follow the classic presentation in many modern textbooks (e.g. Keener & Sneyd, *Mathematical Physiology*, Springer).

Numerical challenges. During the spike, the fast sodium-activation mode and the slower recovery variables create a state-dependent separation of time scales. Explicit Euler may first lose accuracy and, when its absolute-stability restriction is exceeded, can also oscillate, leave the physical gate interval $[0, 1]$, or diverge. An implicit method permits stability at larger steps but is not automatically accurate there. Quantify these effects rather than assuming a fixed step-size advantage. An adaptive controller should take small steps across the stimulus and upstroke and larger steps during quiescent parts of the trajectory.

Your report should address:

- Compute the full 4×4 Jacobian. Report its eigenvalues and a clearly defined time-scale-separation measure at the initial state, at the maximum-upstroke point (maximum dV/dt), and at the voltage peak. Also plot the measure along the trajectory; do not infer it from the rate constants alone.
- Explicit Euler vs. implicit Euler step sizes for the same accuracy: quantify the wall-clock cost.
- Overlay $h\lambda_j(t)$ on the stability regions at the three stated states. Locate experimentally the accuracy threshold and the onset of explicit instability, recording gate-range violations separately.
- Reproduce the single action potential. Report its peak voltage and peak time, and define duration reproducibly (use the time spent above 0 mV, unless you justify another threshold).
- Compare adaptivity with fixed stepping at matched error in V and in (m, h, n) . Report accepted/rejected steps, right-hand-side evaluations, nonlinear iterations, and wall time; conclude from the measurements whether adaptivity cuts the work.

Sanity checks. With $I_{\text{ext}} = 0$, the stated initial state is near the stable resting equilibrium and all four derivatives should be small. For every accurately resolved solution the gates should remain in $[0, 1]$. The prescribed pulse should elicit one action potential followed by recovery toward rest; verify the detailed waveform against a high-accuracy reference you construct yourself (see the project brief, §2 “How to verify without a given answer”). Repeat the reference solve with tighter tolerances and confirm that the reported spike amplitude, peak time, duration definition, and gate extrema are unchanged to the digits claimed.

3. ③ Chemical kinetics: Robertson problem

Phenomenon. A classic benchmark for stiff ODE solvers (H. H. Robertson, 1966). Three species react over widely separated time scales (rate constants from $k_1 = 0.04$ to $k_2 = 3 \times 10^7$, a span of $\sim 10^8$):



These reaction labels are mnemonic. The ODE system below is authoritative and fixes the mass-action coefficient convention; in particular, do not introduce an additional factor of two in the displayed $3 \times 10^7 y_2^2$ term when translating the schematic $B + B$ reaction.

Model. With $y = [A, B, C]$:

$$\begin{aligned} y_1' &= -0.04 y_1 + 10^4 y_2 y_3 \\ y_2' &= 0.04 y_1 - 10^4 y_2 y_3 - 3 \times 10^7 y_2^2 \\ y_3' &= 3 \times 10^7 y_2^2 \\ y(0) &= [1, 0, 0] \end{aligned}$$

The vector field satisfies $(1, 1, 1)^T f(y) = 0$, so the exact trajectory remains on the invariant plane $y_1 + y_2 + y_3 = 1$. Consequently the full 3×3 Jacobian always has a zero eigenvalue. A stiffness ratio that includes that eigenvalue is infinite or undefined; it is not a meaningful diagnostic.

Reproducible protocol. Integrate on $0 \leq t \leq 40$. Include $t = 0$ and at least 200 logarithmically spaced output times from 10^{-8} to 40, while allowing the solver to take its own internal steps. Use the same output times for every method and reference comparison. Construct a tight-tolerance Radau, BDF, or LSODA reference, repeat it with tolerances reduced by a factor of 100, and retain only digits unchanged by that repeat. For uniform-step convergence studies, use at least four successively halved step sizes and state separately whether each run is stable, non-negative, and accurate.

Numerical challenges. Define the reported stiffness ratio using the two nonzero eigenvalues of the full Jacobian,

$$S(t) = \frac{\max_{\lambda_j \neq 0} |\operatorname{Re} \lambda_j(t)|}{\min_{\lambda_j \neq 0} |\operatorname{Re} \lambda_j(t)|},$$

with a stated numerical threshold for classifying an eigenvalue as zero. An equivalent and usually cleaner approach is to eliminate $y_3 = 1 - y_1 - y_2$ and use the two eigenvalues of the reduced Jacobian whenever both are nonzero. The ratio is not informative at the exactly degenerate initial state, so begin the plot at the first positive logarithmic time.

Along a tight-tolerance reference, $|\lambda|_{\max}$ reaches approximately 3.39×10^3 by $t = 40$ (not 3×10^7 , because $y_2 = O(10^{-5})$). For orientation, the nonzero-eigenvalue ratio is about 3.0×10^3 at $t = 10^{-4}$, 5.4×10^3 at $t = 10^{-2}$, and 1.58×10^5 at $t = 40$. These are check values, not substitutes for the student's own computation. The most restrictive negative-real eigenvalue suggests an explicit-Euler bound near $h < 2/(3.39 \times 10^3) \approx 5.9 \times 10^{-4}$, but the state-dependent bound, accuracy, and positivity must all be tested. This is a local frozen-Jacobian estimate, not a sufficient nonlinear stability theorem. The Jacobian varies along the trajectory, and eigenvalues alone can be misleading for non-normal systems; compare the estimate with an empirical step sweep and report any discrepancy. You will need a high-accuracy solution to check against — construct it yourself (see the project brief, §2 “How to verify without a given answer”).

Your report should address:

- Plot both nonzero Jacobian eigenvalues and the defined stiffness ratio against logarithmic time. Explain the structural zero eigenvalue and the treatment of the initial degeneracy.
- Show where explicit Euler becomes unstable, inaccurate, or negative. Compare implicit Euler at matched error; “it did not blow up” is not an accuracy claim.
- Track conservation $y_1 + y_2 + y_3 = 1$, componentwise non-negativity, and full-state error as separate diagnostics. Conservation alone is weak: inaccurate or negative solutions can still conserve the sum.
- The final value of y_2 reported by your implicit method at several tolerances, cross-checked against your own high-accuracy reference.

Sanity checks. A consistently implemented Runge–Kutta method preserves the linear invariant algebraically when its stage equations are solved exactly. For an implicit method, an inexact nonlinear solve produces an invariant defect controlled by the nonlinear residual; Newton need not be driven to machine precision. Choose its stopping tolerance small enough that algebraic error is negligible relative to the requested time-discretisation error, then demonstrate this with a tolerance sweep. At $t = 40$, an independent reference gives roughly $y = (0.7158271, 9.1855 \times 10^{-6}, 0.2841637)$; report only the digits confirmed by your own tightened reference. After its early maximum, y_2 should relax without spurious oscillation, all components should remain non-negative, and their sum should remain one.

4. ④ Financial SDE: GBM benchmark and non-affine volatility

Phenomenon. In finance, asset prices are modelled with stochastic differential equations. The simplest is geometric Brownian motion (GBM):

$$dS = \mu S dt + \sigma S dW$$

with solution $S(t) = S_0 \exp((\mu - \sigma^2/2)t + \sigma W(t))$ — an *exact* formula exists, making this an ideal validation problem.

Model. The standard finance discretisation is in the log price $X_t = \log S_t$: Itô’s lemma gives the constant-coefficient SDE

$$dX = (\mu - \frac{1}{2}\sigma^2) dt + \sigma dW,$$

so Euler–Maruyama on X , $X_{n+1} = X_n + (\mu - \sigma^2/2)\Delta t + \sigma \Delta W_n$, reproduces the exact Gaussian transition and $S_n = e^{X_n}$ stays positive — this is the exact-simulation formula. But on X the noise is additive: the Milstein correction vanishes, so one cannot *measure* a strong order of 1 there. To demonstrate convergence order, discretise the price itself, where the noise is state-dependent:

$$S_{n+1} = S_n + \mu S_n \Delta t + \sigma S_n \Delta W_n \quad (\text{EM})$$

$$S_{n+1} = S_n + \mu S_n \Delta t + \sigma S_n \Delta W_n + \frac{1}{2} \sigma^2 S_n (\Delta W_n^2 - \Delta t) \quad (\text{Milstein})$$

Numerical challenges. This is an *SDE*, so the ODE stability/order theory must be replaced by strong vs. weak convergence. Great for showing students that order concepts carry over but with fractional orders.

Your report should address:

- Strong convergence: $E|S_N - S(T)|$ vs. Δt on log-log — slope $\frac{1}{2}$ for EM, 1 for Milstein.
- Weak convergence of $E[S(T)]$ — expected slope 1 for both methods. Use independent batches or confidence intervals, and choose enough paths that Monte Carlo error does not hide the time-stepping bias.
- The distribution of $S(T)$ vs. the exact lognormal law (histogram overlay, quantile comparison).

- *Optional (☆☆☆):* positivity — determine when an EM or Milstein update can produce a negative value. Neither scheme is unconditionally positivity-preserving. Measure how often negativity occurs as Δt varies and explain the possible effect on option prices.

Sanity check. $E[S(T)] = S_0 e^{\mu T}$ exactly; Var matches the lognormal formula.

But why simulate a model that has a formula? It is a fair objection, and answering it is the real lesson. GBM is a *rare gift*: its transition has a closed form, so $S(t)$ can be simulated *exactly* via $S_{n+1} = S_n \exp((\mu - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t} Z_n)$ — no discretisation error at all, so Euler–Maruyama is unnecessary there. For a model in which time discretisation is genuinely necessary, use the following **non-affine stochastic-volatility model**. Write $X_t = \log S_t$ and let

$$\begin{aligned} dX_t &= \left(\mu - \frac{1}{2}g(Y_t)^2\right) dt + g(Y_t) dW_t^{(1)}, \\ dY_t &= \kappa(\theta - Y_t)dt + \xi\sqrt{1 + Y_t^2} dW_t^{(2)}, \\ g(y) &= \sigma_{\min} + \frac{\sigma_{\max} - \sigma_{\min}}{1 + e^{-y}}, \quad d\langle W^{(1)}, W^{(2)} \rangle_t = \rho dt, \end{aligned}$$

where $0 < \sigma_{\min} < \sigma_{\max}$, $\kappa, \xi > 0$, and $|\rho| < 1$. The bounded logistic function keeps instantaneous asset volatility positive, while the nonlinear volatility-factor diffusion makes the joint process non-affine. No closed-form transition distribution or practical exact sampler is known for this generic model, so numerical SDE integration is part of the model solution rather than merely a comparison with a known formula.

Use the reproducible benchmark

$$\begin{aligned} S_0 = 100, \quad Y_0 = 0, \quad \mu = 0.05, \quad \kappa = 2, \quad \theta = -0.2, \quad \xi = 0.6, \quad \rho = -0.7, \\ \sigma_{\min} = 0.10, \quad \sigma_{\max} = 0.50, \quad T = 1. \end{aligned}$$

Generate correlated increments from independent $Z_{1,n}, Z_{2,n} \sim N(0, 1)$:

$$\Delta W_n^{(1)} = \sqrt{\Delta t} Z_{1,n}, \quad \Delta W_n^{(2)} = \sqrt{\Delta t}(\rho Z_{1,n} + \sqrt{1 - \rho^2} Z_{2,n}).$$

Apply Euler–Maruyama to (X, Y) and recover $S_n = e^{X_n}$; this representation preserves $S_n > 0$ automatically. Do not quote the scalar Milstein order-one result for this two-noise system: a full multidimensional Milstein method may require cross terms and simulated iterated stochastic integrals.

Recommended extension for a finance student:

- Couple every coarse path to a much finer EM reference path by summing the fine Brownian increments. Plot $E|S_T^{(h)} - S_T^{(h_{\text{ref}})}|$ with confidence intervals; describe the observed strong rate as empirical, because the reference is numerical rather than exact.
- For weak convergence, estimate a payoff such as $E[(S_T - 100)^+]$. Use the same fine reference, common random numbers, and confidence intervals. Do not claim a slope once estimated bias is comparable to its sampling uncertainty.
- Check Brownian-increment sample variances and correlation, convergence under refinement, finiteness of all paths, and positivity of $S = e^X$. Repeat with at least two finer reference steps to show that the benchmark itself is sufficiently resolved.
- Compare this experiment with GBM: GBM supplies an exact pathwise benchmark, whereas the non-affine model requires a carefully verified numerical reference. State clearly that agreement between two discretised solutions is evidence, not an exact-error calculation.

5. ⑤ Matrix differential equations: polar-factor gradient flow (extension topic)

Phenomenon. Matrix ODEs arise as continuous-time versions of optimisation algorithms. Here the goal is to deform a nonsingular square matrix continuously toward the orthogonal factor in its polar decomposition. This gives a nonlinear matrix problem with an exact solution against which the time integrators can be tested.

Model. Let $X(t) \in \mathbb{R}^{n \times n}$ and assume that X_0 is nonsingular. (In a generic IVP code, the state is $y = \text{vec}(X)$.) Define

$$\Phi(X) = \frac{1}{4} \|X^T X - I\|_F^2, \quad \frac{dX}{dt} = -\nabla \Phi(X) = X(I - X^T X), \quad X(0) = X_0.$$

If $X_0 = U \text{diag}(s_{1,0}, \dots, s_{n,0}) V^T$ is an SVD with $s_{i,0} > 0$, then the singular vectors remain fixed and

$$X(t) = U \text{diag}(s_1(t), \dots, s_n(t)) V^T, \quad s_i(t) = \frac{s_{i,0}}{\sqrt{s_{i,0}^2 + (1 - s_{i,0}^2)e^{-2t}}}.$$

Thus singular values below 1 increase, those above 1 decrease, their ordering is preserved, and $X(t) \rightarrow UV^T$, the orthogonal polar factor. The nonsingularity assumption matters: a zero singular value remains zero, so a rank-deficient initial matrix cannot converge to an orthogonal matrix.

This is a *dissipative* flow, not a conservative one. Its Lyapunov function satisfies

$$\frac{d}{dt} \Phi(X(t)) = -\|X(I - X^T X)\|_F^2 \leq 0.$$

Also distinguish this ambient-space flow from constrained optimisation on the orthogonal group. If $X^T X = I$ and $G = \nabla f(X)$, the tangent projection is

$$P_X(G) = G - X \text{sym}(X^T G) = X \text{skew}(X^T G),$$

so the corresponding Riemannian gradient flow is $X' = -P_X(G)$. The expression $(I - XX^T)G$ is not the required tangent projection (and is identically zero for square orthogonal X).

Numerical challenges. Large initial singular values create a rapid transient and can impose a severe explicit-step restriction. At an orthogonal equilibrium, tangent perturbations are neutral because every orthogonal matrix is an equilibrium, whereas normal perturbations decay at rate 2. Useful diagnostics are the exact global error, $\|X^T X - I\|_F$, $\Phi(X)$, the singular values, and the defect in the Lyapunov identity. Do not expect $\|X^T X - I\|_F$ to be round-off size at a finite final time: compare it with the nonzero exact value given by the formula above.

Your report should address:

- Vectorise the matrix ODE and run all three required methods. Verify their observed orders using the full-matrix exact solution above.
- Use the reproducible benchmark

$$X_0 = R(0.4) \text{diag}(0.2, 1.4, 3) R(-0.7)^T, \quad R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and state the interval, tolerances, and step sizes used.

- Check whether the computed singular values move toward 1 without crossing, and whether the numerical $\Phi(X)$ is monotone. Explain why a method need not preserve either property when the step is too large.
- Linearise about an orthogonal equilibrium and identify the neutral tangent directions and contracting normal directions. Relate this analysis, and the rapid transient from $s_{3,0} = 3$, to observed stability limits.
- Repeat one experiment with a rank-deficient X_0 and verify that the zero singular value remains zero; explain why the limit is then only a partial isometry.

Sanity check. For diagonal $X_0 = \text{diag}(x_i)$ with $x_i > 0$, the matrix remains diagonal and each entry is given by the same scalar formula as $s_i(t)$ above. For general nonsingular X_0 , reconstruct the exact matrix from its initial SVD. Order tests must compare the numerical solution with this finite-time exact matrix, rather than with the limiting polar factor UV^T .

Optional application for Dimension B: microwave-filter topology transformation

This extension is not required. A sound, clearly explained and verified investigation can provide evidence for **Assessment Dimension B — Depth and Novelty**; merely adding the example or producing an extra plot does not by itself earn a particular level.

A coupled-resonator microwave filter can be represented by a real symmetric *coupling matrix* M_0 . Its off-diagonal entries describe couplings between resonators, so the zero pattern represents the physical network topology. A matrix obtained during filter synthesis may have the desired frequency response but an inconvenient, densely connected topology. The engineering task is to find an equivalent matrix with prescribed couplings removed.

Let $Q \in O(n)$ and set

$$M(Q) = Q^T M_0 Q.$$

In a model that includes source and load nodes, restrict $Q = \text{diag}(1, Q_r, 1)$ so that only the resonator coordinates are rotated; this restriction is essential for preserving the port response. Orthogonal similarity preserves the eigenvalues of M_0 , while the change of basis alters its coupling pattern. Given a symmetric binary mask W , with $W_{ij} = 1$ for a coupling that should vanish, minimise

$$J(Q) = \frac{1}{2} \|W \circ M(Q)\|_F^2, \quad Q \in O(n),$$

where $\circ$ denotes entrywise multiplication. With $S = W \circ M$, let Π_r retain only the resonator–resonator block of a matrix and set its source/load rows and columns to zero. One negative-gradient formulation on the port-fixing subgroup is

$$Q' = Q\Omega, \quad \Omega = \Pi_r([S, M]), \quad [S, M] = SM - MS, \quad M' = [M, \Omega].$$

Because M and S are symmetric, both $[S, M]$ and Ω are skew-symmetric; hence the exact flow preserves $Q^T Q = I$, fixes the port coordinates, and is isospectral. This is a continuous analogue of the Givens-rotation similarity transformations used in practical coupling- matrix synthesis.

Published numerical benchmark. Use the four-resonator transversal-to- folded example of Pflüger, Waldschmidt and Ziegler (2016, Fig. 1 and Eq. (19)). In the coordinate order (S, 1, 2, 3, 4, L) (source, four resonators, load), their initial normalized coupling matrix is

$$M_0 = \begin{pmatrix} 0 & 0.3641 & -0.6537 & 0.6677 & -0.3433 & 0.0151 \\ 0.3641 & 1.3142 & 0 & 0 & 0 & 0.3641 \\ -0.6537 & 0 & 0.7831 & 0 & 0 & 0.6537 \\ 0.6677 & 0 & 0 & -0.8041 & 0 & 0.6677 \\ -0.3433 & 0 & 0 & 0 & -1.2968 & 0.3433 \\ 0.0151 & 0.3641 & 0.6537 & 0.6677 & 0.3433 & 0 \end{pmatrix}.$$

The paper’s structure matrix uses 0 for a forbidden entry and 1 for an allowed entry. To match the convention

used in J above, use its off-diagonal complement as the mask

$$W = \begin{pmatrix} 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{pmatrix}.$$

Thus $W_{ij} = 1$ marks an entry to suppress and diagonal entries are not penalised. Keep $Q = \text{diag}(1, Q_r, 1)$ with $Q_r \in O(4)$. The numbers above are transcribed to the four decimal places printed in the paper, so use a tolerance appropriate to rounded data rather than requiring J to reach machine zero. Symmetrise computed matrices before an eigensolver only to remove round-off asymmetry, not as part of the evolution.

For a Dimension-B investigation, use a supplied or cited coupling matrix and topology mask, and compare a standard vectorised integrator with at least one manifold-aware update (for example, an exponential step or a QR/polar retraction). Report $J(Q)$, the orthogonality defect $\|Q^T Q - I\|_F$, eigenvalue drift in M , and the unwanted entries before and after transformation. Discuss step-size sensitivity, convergence to different equivalent topologies from different initial rotations, or the residual obtained when the requested topology is not exactly attainable. State clearly which part goes beyond the required work and why it adds mathematical or algorithmic depth.

Starting reference. S. Pflüger, C. Waldschmidt and V. Ziegler, “Coupling Matrix Extraction and Reconfiguration Using a Generalized Isospectral Flow Method,” *IEEE Transactions on Microwave Theory and Techniques*, 64(1), 148–159, 2016, <https://doi.org/10.1109/TMTT.2015.2498188>.

Choosing your topic

Consideration	Suggests
Your team likes mechanics/physics	① or ②
Your team likes stiff-method mathematics	③
Your team likes finance/probability	④
Your team likes linear algebra / optimization	⑤
You want the cleanest validation story	① (energy) or ④ (exact solution)
You want the most dramatic “explicit fails” demo	② or ③

Whichever you choose, the ideas of scaffolding code, stability analysis, and adaptive controller support all five — only the right-hand side $f(t, y)$ changes.

Further inspiration (research links)

- **Chain flying with a car** — rigid-body / flexible-body dynamics problems; natural extensions of topic ①. The repo’s Flying Chain/ sub-project is a worked example.
- **Swimming robot (fluid–rigid-body coupling)** — a unified fluid-robot multiphysics framework from Carnegie Mellon / MIT (RSS 2026): <https://unified-fluid-robot-multiphysics.github.io/>. This is a heavier *multiphysics* problem (fluid + rigid body), closer to the PDE projects than to the ODE IVP one, but the *mechanism design* (variational integrator, energy diagnostics) is a great motivation for topic ①.